

Object-Oriented Modelling and Programming in Engineering

Homework 1

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From an oscillating energy system, the equation for the power:

$$P(t) = a1 * t + \sin(2 * \pi * f1 * t)$$

$$a1 = 125353/40\,000$$

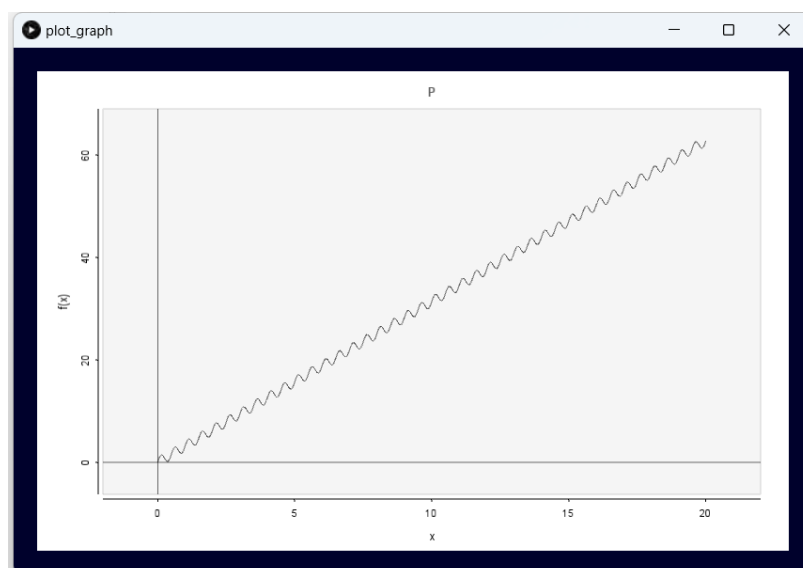
$$= 3.1338\text{W}$$

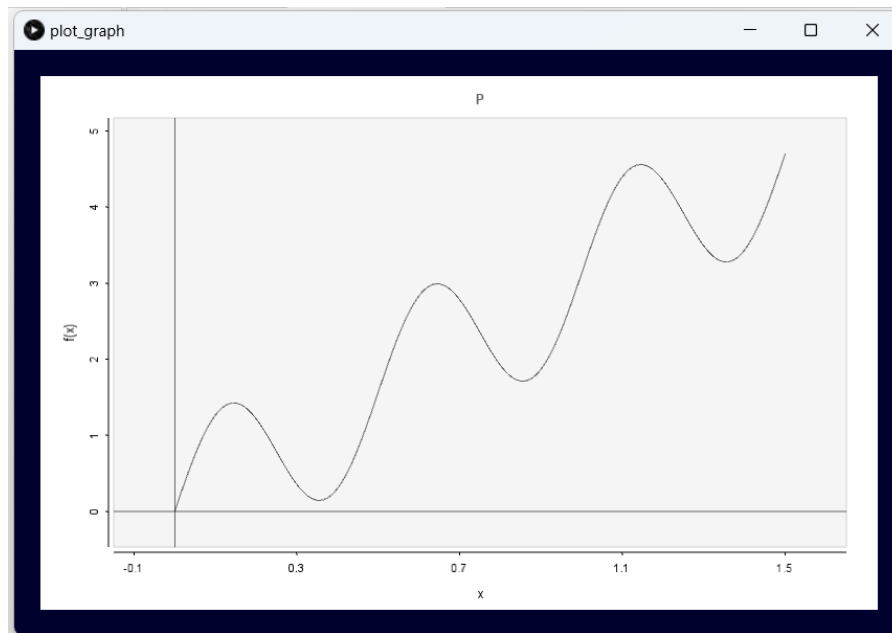
$$f1 = 2\text{Hz}$$

After integrating the energy consumption $E(t)$ for $t=20$ seconds will be –

$$E(t) = 626.765$$

1. **Exact result of the integration is-**
626.765
2. **Plot of the function $P(t)$ is represented below-**





3. Nassi-Schneiderman diagrams for all the three algorithms-

1. Integral function for method 1-

```

integral1():double
Function F = new Function("P(t) = ((125353/40000)*t) + sin(2*pi*2*t)");
double k, value=0;
for (k=0;k<=20;k=k+0.1)
    value = value+ (F.calculate(k)*0.1);
return value

```

2. Integral function for method 2-

```

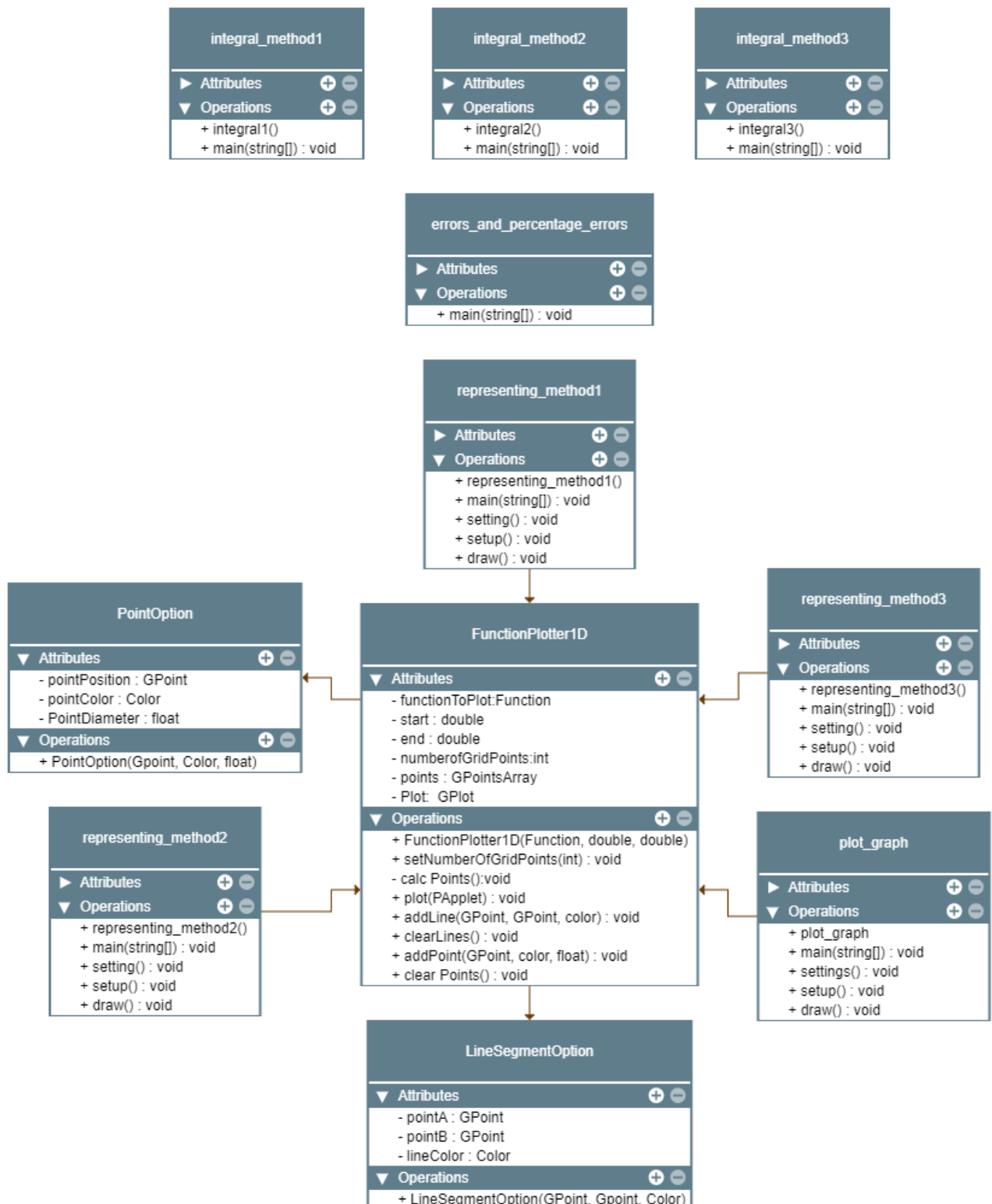
integral2():double
Function F = new Function("P(t) = ((125353/40000)*t) + sin(2*pi*2*t)");
double i, value=0;
for (i=0;i<=20;i=i+0.1)
    value = value+ (((F.calculate(i)+F.calculate(i+0.1))/2)*0.1);
return value

```

3. Integral function for method 3-

integral3():double	
Function F = new Function("P(t) = ((125353/40000)*t) + sin(2*pi*2*t)");	
double j, value=0;	
for (j=0;j<=20;j=j+0.1)	
	double a=((F.calculate(j+0.1)-F.calculate(j))/(0.1));
	double b=(F.calculate(j)-j*a);
	double c= (a*Math.pow(j+0.1, 2)/2)+(F.calculate(j)-a*j)*(j+0.1);
	double d= (a*Math.pow(j, 2)/2)+(F.calculate(j)-a*j)*j);
	value=value+(c-d);
return value	

4. UML-diagrams for the software structure-



5. Results of the numerical integration-

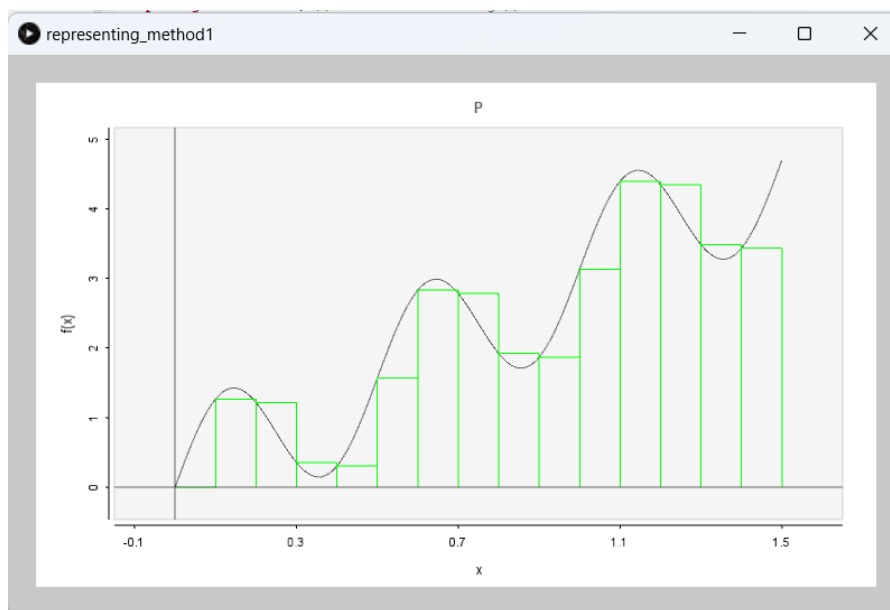
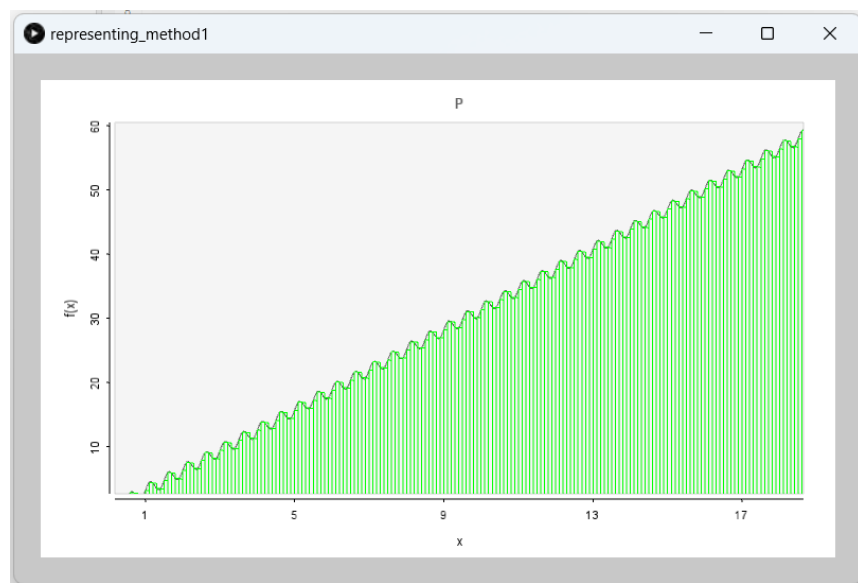
Integration value by method 1- 623.6311749999994

Integration value by method 2- 626.7649999999999

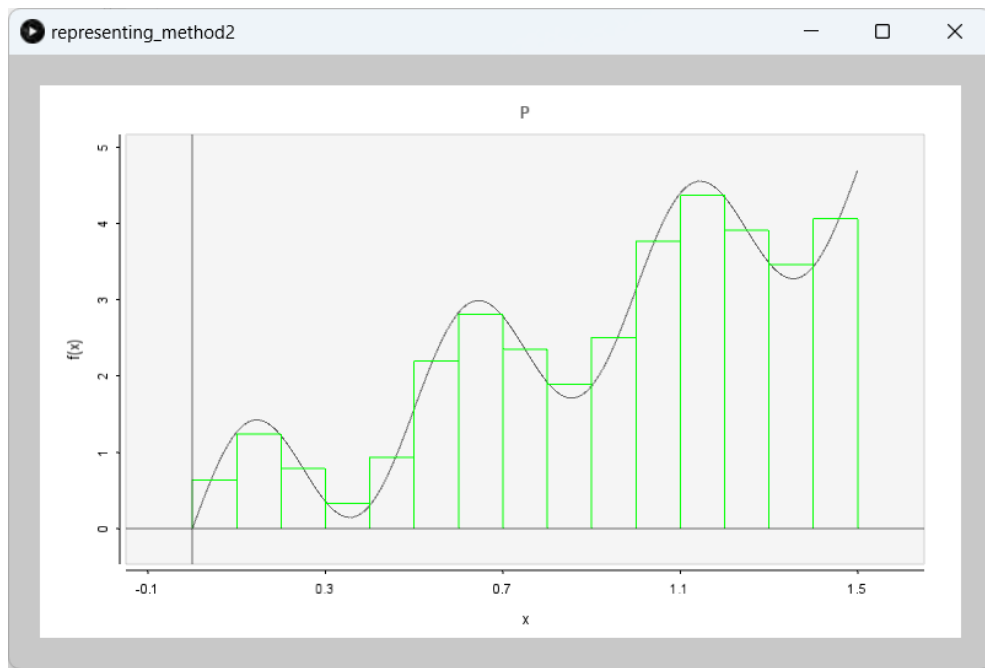
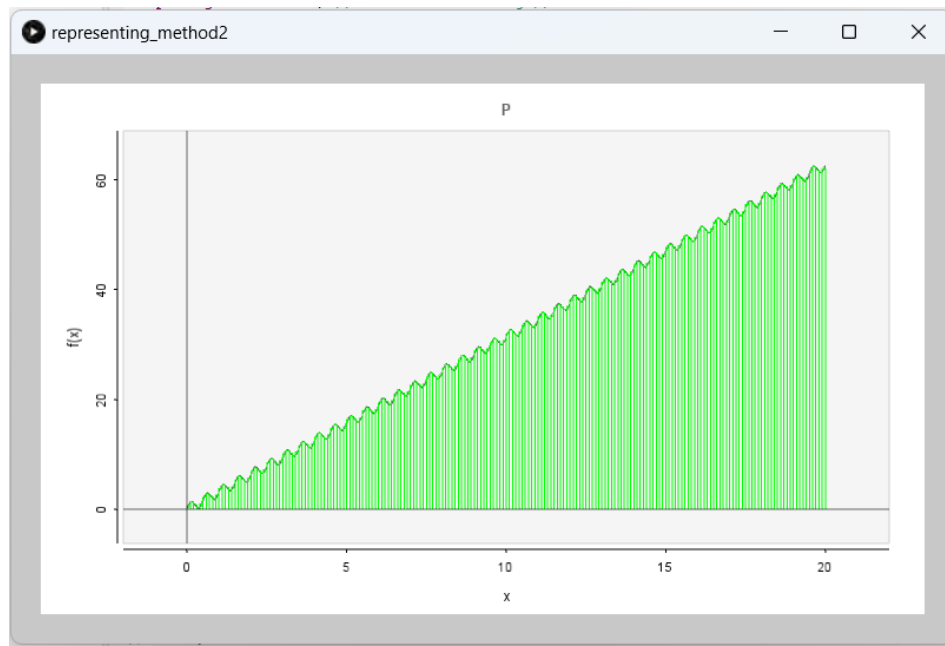
Integration value by method 3- 626.7650000000016

6. Visualization of the integral –

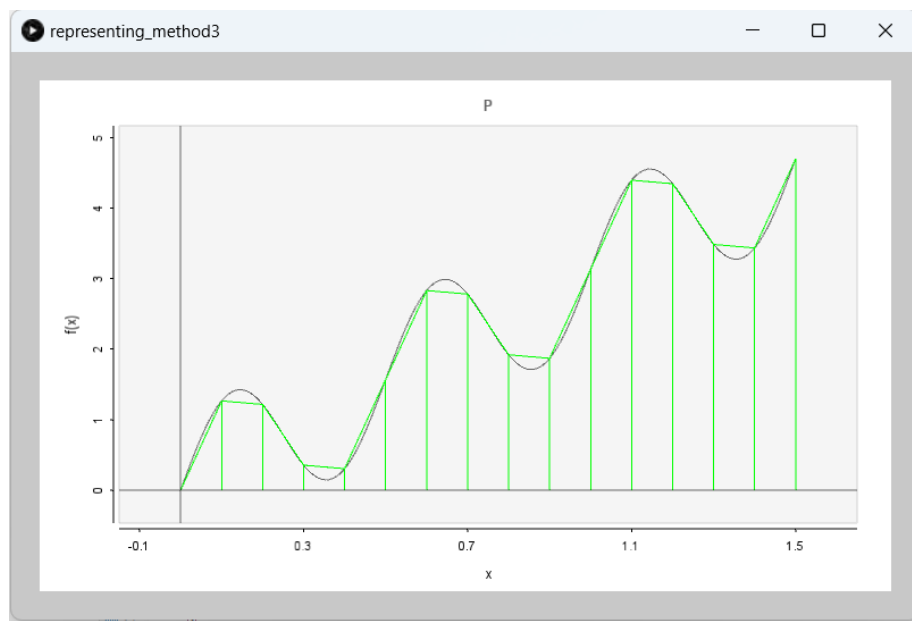
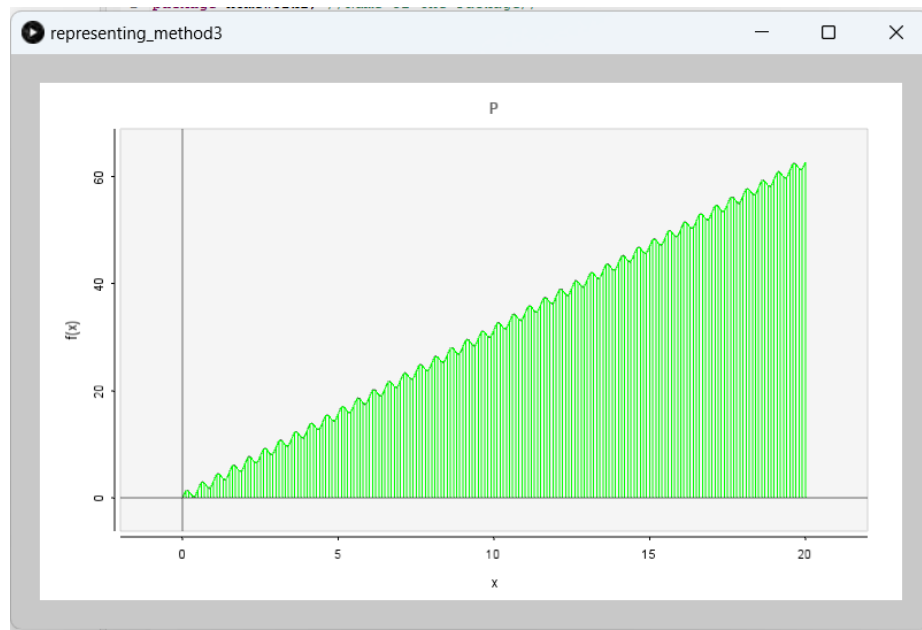
Visualization of the integral by method 1-



Visualization of the integral by method 2-



Visualization of the integral by method 3-



7. Absolute and relative errors of the numerical results-

Integral value for method 1 is: 623.6311749999994

Absolute error for method 1 is: 3.133825000000057

Absolute error % for method 1 is: 313.3825000000057

Relative error for method 1 is: 0.005000000000000091

Relative error % for method 1 is: 0.5000000000000091

Integral value for method 2 is: 626.7649999999999

Absolute error for method 2 is: 1.0231815394945443E-12

Absolute error % for method 2 is: 1.0231815394945443E-10

Relative error for method 2 is: 1.6324803387147405E-15

Relative error % for method 2 is: 1.6324803387147406E-13

Integral value for method 3 is: 626.76500000000016

Absolute error for method 3 is: 1.5916157281026244E-12

Absolute error % for method 3 is: 1.5916157281026244E-10

Relative error for method 3 is: 2.5394138602229294E-15

Relative error % for method 3 is: 2.5394138602229295E-13
